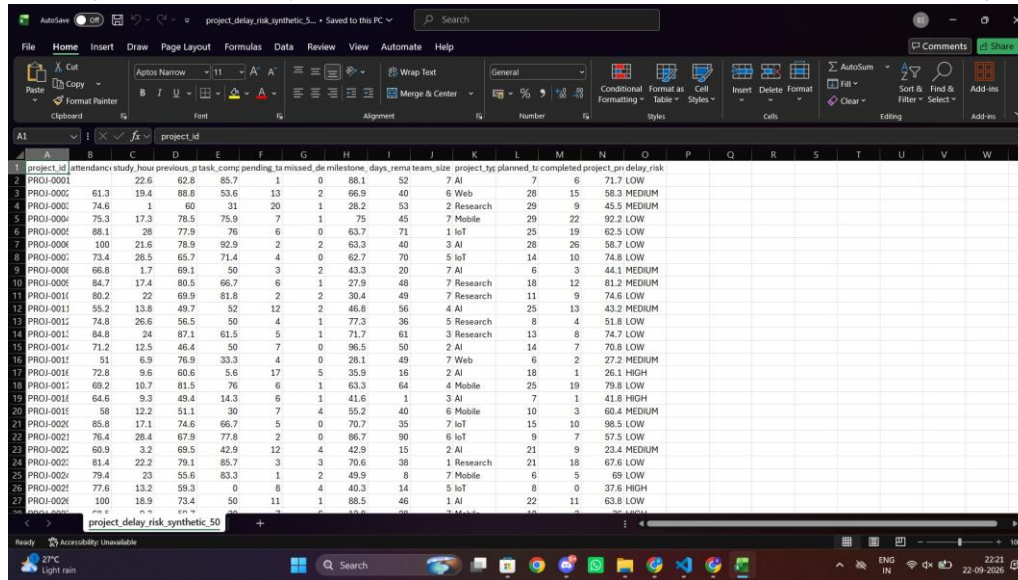


AI-AcadPredict – AI Service Results Summary

AI-Integrated Academic Project Management and Delay Risk Prediction Platform

1. AI/ML Model Development

- Generated a synthetic dataset containing 5,000 academic project records.
- The target variable is delay_risk with three classes: LOW, MEDIUM, and HIGH.
- A random seed of 42 was used for reproducible results.
- Missing values were injected into selected numerical columns for preprocessing validation.



The screenshot displays a Microsoft Excel spreadsheet titled "project_delay_risk_synthetic_50". The spreadsheet contains a table with columns labeled A through W. The first column (A) contains project IDs (e.g., PROJ-0001, PROJ-0002). The second column (B) contains attendance values. The third column (C) contains study values. The fourth column (D) contains hour values. The fifth column (E) contains previous values. The sixth column (F) contains p values. The seventh column (G) contains task values. The eighth column (H) contains comp values. The ninth column (I) contains pending values. The tenth column (J) contains ts values. The eleventh column (K) contains missed values. The twelfth column (L) contains de values. The thirteenth column (M) contains milestone values. The fourteenth column (N) contains days values. The fifteenth column (O) contains rema values. The sixteenth column (P) contains team values. The seventeenth column (Q) contains size values. The eighteenth column (R) contains project values. The nineteenth column (S) contains hy values. The twentieth column (T) contains planned values. The twenty-first column (U) contains ts values. The twenty-second column (V) contains completed values. The twenty-third column (W) contains project values. The twenty-fourth column (X) contains pri values. The twenty-fifth column (Y) contains delay values. The twenty-sixth column (Z) contains risk values. The data is organized into rows, with the first row being a header and subsequent rows containing numerical and categorical values. The delay_risk column shows values like LOW, MEDIUM, and HIGH.

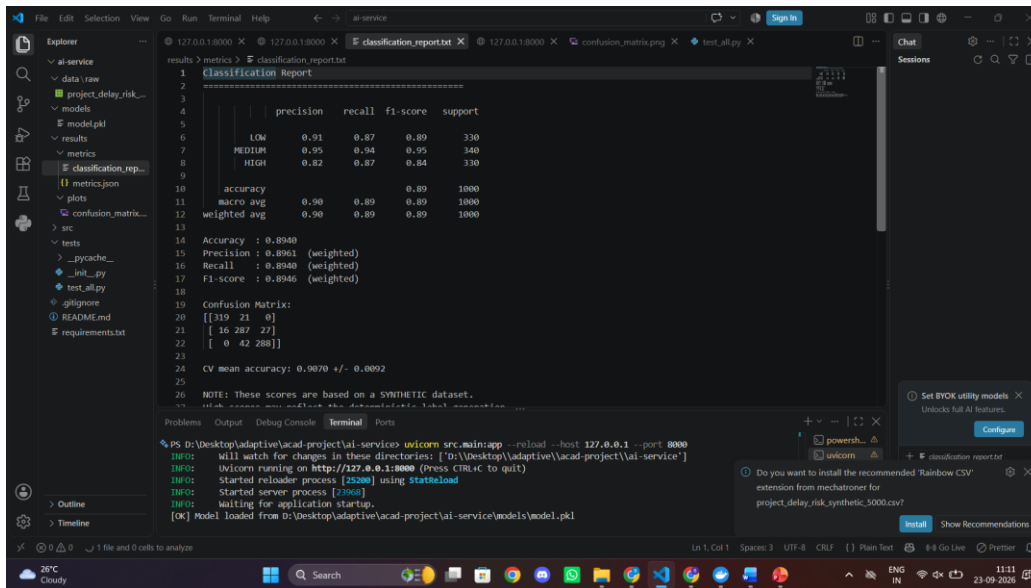
2. Data Preprocessing

- Used a unified Scikit-learn Pipeline to reduce data leakage.
- Applied median imputation to numerical features.
- Applied categorical imputation and OneHotEncoder to categorical features.
- Configured the encoder with handle_unknown='ignore'.

3. Model and Evaluation

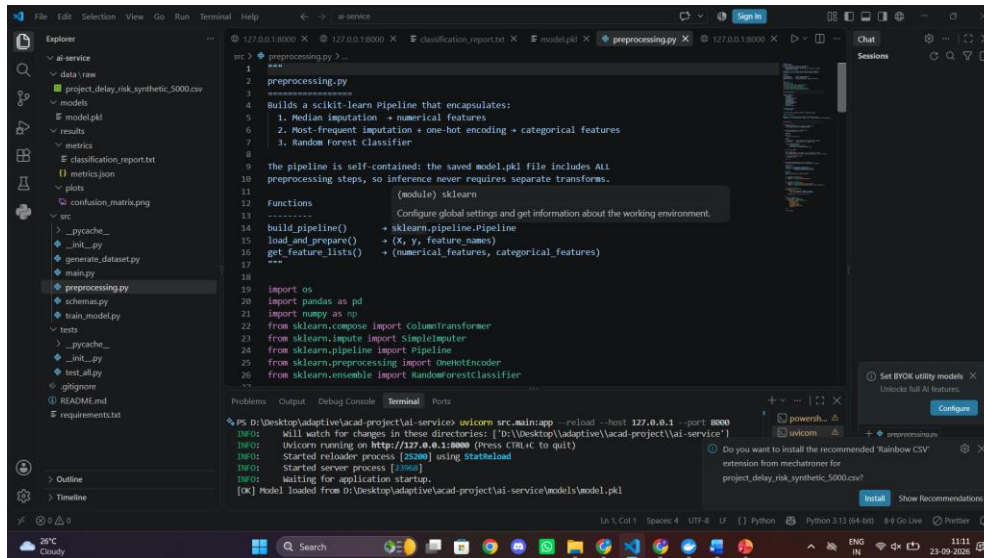
- Used a Random Forest Classifier with 100 estimators, maximum depth 12, minimum split size 5, and random state 42.
- Used a stratified 80:20 train-test split: 4,000 training records and 1,000 testing records.
- Used five-fold cross-validation to assess consistency.

Evaluation Metric	Result
Test Accuracy	89.40%
5-Fold CV Accuracy	90.70% ± 0.92%
Weighted Precision	89.61%
Weighted Recall	89.40%
Weighted F1-Score	89.46%



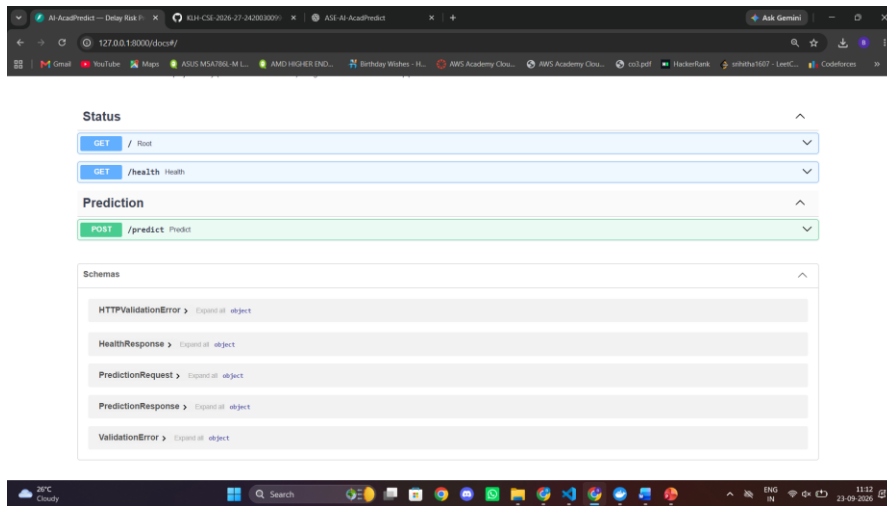
4. Generated Artifacts

- metrics.json – stores model evaluation metrics.
- confusion_matrix.png – shows prediction performance by risk class.
- classification_report.txt – contains classification metrics.
- models/model.pkl – saved trained model pipeline.



5. FastAPI AI Service

- Implemented the AI prediction service using FastAPI.
- Main files: src/main.py and src/schemas.py.
- Used Pydantic v2 for request validation and schema mapping.
- Supported camelCase fields such as studyHours, taskCompletion, and daysRemaining.
- Loaded the trained model using Joblib.



Endpoint	Method	Purpose
/	GET	Basic service response
/health	GET	Service health check
/predict	POST	Predicts project delay risk
/docs	GET	Interactive Swagger documentation

6. API Results

- The FastAPI server was started and tested successfully.
- A sample project input returned MEDIUM risk with a probability of 0.8781.
- Other test inputs returned LOW, MEDIUM, and HIGH risk predictions.
- The probability is the model's confidence estimate for an individual prediction, not the overall model accuracy.

Predicted Risk	Probability	Description
MEDIUM	0.8781	Sample project input
LOW	0.9883	High-confidence low-risk result
LOW	0.9628	Low-risk result
HIGH	0.9980	High-confidence high-risk result
MEDIUM	0.7146	Medium-risk result
LOW	0.9960	High-confidence low-risk result

AI-AcadPredict - Delay Risk P

127.0.0.1:8000/docs/Prediction/predict_post

Responses

Out

```
curl -X POST \
  http://127.0.0.1:8000/predict \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "attitude": 45,
    "prelaunchPerformance": 40,
    "launchPerformance": 20,
    "pendingIndex": 12,
    "prelaunchIndex": 5,
    "missionCompletion": 20,
    "launchIndex": 5,
    "mission": 2,
    "projectType": "AI",
    "projectProgress": 10
  }'
```

Request URL

http://127.0.0.1:8000/predict

Server response

Code Details

200

Response body

```
{
  "risk": "low",
  "probability": 0.999
}
```

Response headers

```
content-length: 35
content-type: application/json
date: Tue, 22 Sep 2026 10:48:19 GMT
server: waitress
```

Responses

Code Description

Links

ASL-AI-AcadPredict

127.0.0.1:8000/docs/Prediction/predict_post

Execute Clear

Responses

Out

```
curl -X POST \
  http://127.0.0.1:8000/predict \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "attitude": 50,
    "prelaunchPerformance": 50,
    "launchPerformance": 40,
    "pendingIndex": 4,
    "prelaunchIndex": 2,
    "missionCompletion": 20,
    "launchIndex": 2,
    "mission": "Research",
    "projectType": "Research",
    "projectProgress": 50
  }'
```

Request URL

http://127.0.0.1:8000/predict

Server response

Code Details

200

Response body

```
{
  "risk": "medium",
  "probability": 0.750
}
```

Response headers

```
content-length: 58
content-type: application/json
date: Tue, 22 Sep 2026 10:49:59 GMT
server: waitress
```

Responses

Code Description

Links

AI-AcadPredict - Delay Risk P

127.0.0.1:8000/docs/Prediction/predict_post

Responses

Out

```
curl -X POST \
  http://127.0.0.1:8000/predict \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "attitude": 50,
    "prelaunchPerformance": 50,
    "launchPerformance": 40,
    "pendingIndex": 2,
    "prelaunchIndex": 4,
    "missionCompletion": 50,
    "launchIndex": 4,
    "mission": "AI",
    "projectType": "AI",
    "projectProgress": 50
  }'
```

Request URL

http://127.0.0.1:8000/predict

Server response

Code Details

200

Response body

```
{
  "risk": "low",
  "probability": 0.999
}
```

Response headers

```
content-length: 35
content-type: application/json
date: Tue, 22 Sep 2026 10:50:59 GMT
server: waitress
```

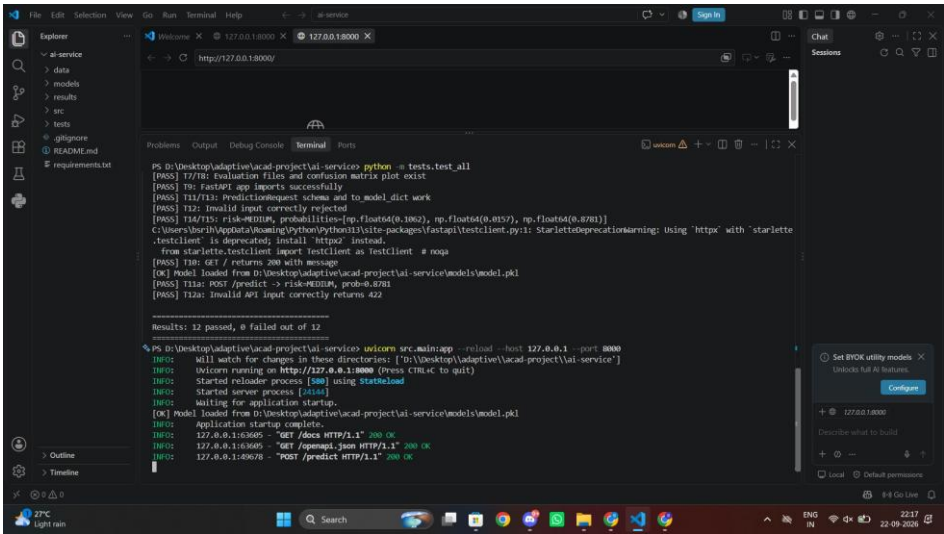
Responses

Code Description

Links

7. Automated Testing

- All 12 automated tests passed successfully.
- Testing covered dataset generation, missing values, model loading, evaluation artifacts, FastAPI importability, schema validation, pipeline inference, and API responses.
- The /predict endpoint returned HTTP 200 for valid input and HTTP 422 for invalid input.



The screenshot shows a VS Code terminal window with the following content:

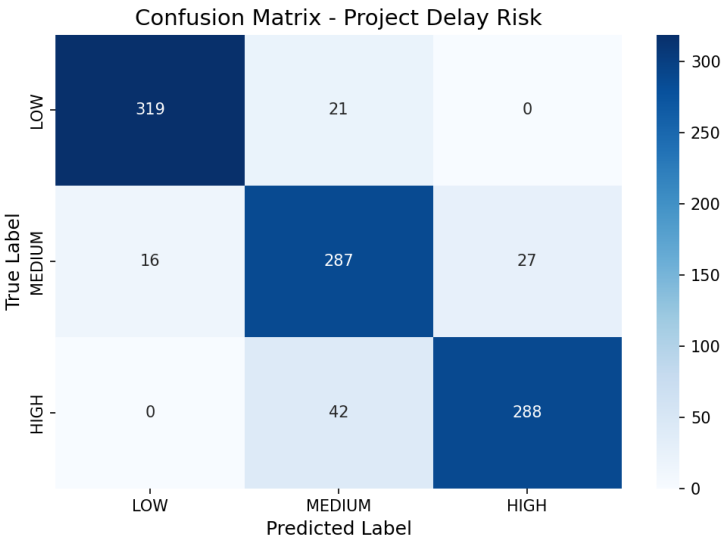
```
PS D:\Desktop\adaptive\acad-project\ai-service> python -m tests.test_all
[INFO] 17/18: Evaluation files and confusion matrix plot exist
[INFO] 19: FastAPI app imports successfully
[INFO] 111/111: Prediction request schema and to model dict work
[INFO] 112: Invalid input correctly rejected
[INFO] 114/115: risk-estimate, probabilities=[p.float64(0.1002), m.float64(0.4037), n.float64(0.4961)]
C:\Users\borch\AppData\Local\Programs\Python\Python311\Scripts\fastapi\testclient.py:1: StarletteDeprecationWarning: Using 'https' with 'starlette.testclient' is deprecated; install 'httpx' instead.
[INFO] 116: GET / returns 200 with message
[INFO] 117: Model loaded from D:\Desktop\adaptive\acad-project\ai-service\models\model.pkl
[INFO] 118: POST /predict -> risk-estimate, probs=[0.4961, 0.4037]
[INFO] 119: Invalid API input correctly returns 422

Results: 12 passed, 0 failed out of 12

PS D:\Desktop\adaptive\acad-project\ai-service> uvicorn src.main:app --reload --host 127.0.0.1 --port 8000
[INFO] Will watch for changes in these directories: ['D:\Desktop\adaptive\acad-project\ai-service']
[INFO] Uvicorn running on http://127.0.0.1:8000 (press CTRL+C to quit)
[INFO] Started reload process [500] using Stateload
[INFO] Started server process [1166]
[INFO] Waiting for application startup.
[INFO] Model loaded from D:\Desktop\adaptive\acad-project\ai-service\models\model.pkl
[INFO] Application startup complete.
[INFO] 127.0.0.1:63605 - "GET /docs HTTP/1.1" 200 OK
[INFO] 127.0.0.1:63605 - "GET /openapi.json HTTP/1.1" 200 OK
[INFO] 127.0.0.1:63606 - "POST /predict HTTP/1.1" 200 OK
```

8. Overall Result

The AI service is operational and successfully predicts academic project delay risk through a FastAPI endpoint. The Random Forest model was trained, evaluated, saved, and connected to the API. This implementation demonstrates the AI/ML component of AI-AcadPredict and can be integrated with the main backend and frontend.



9. Future Improvements

- Integrate the AI service with the complete backend and frontend.
- Validate the model using real academic project data.
- Add adaptive reminders and risk-based alerts.
- Deploy the FastAPI service for remote access.
- Create dashboards for project progress and risk visualization.